MANIKANTA DARAPUREDDY

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SUMMARY

AI/ML Engineer specializing in **real-time intelligent systems** using **LLMs, RAG pipelines**, and **computer vision**. Built **scalable AI applications** with **sub-second latency**, achieving up to **99% accuracy** while reducing **infrastructure costs**. Focused on **production-ready architectures** and **end-to-end deployment**.

EDUCATION

- **B. Tech** in Computer Science and Engineering (**AI & Data Science**)
  Institute: BVCE | Year: 2022–2026 | CGPA: 8.87
- **Intermediate** in **MPC**
  Institute: Educare Junior College | Year: 2020–2022 | GPA: 7.1
- **SSC**- Educare E.M School | Year: 2020 | GPA: 9.3

Technologies & Skills

Languages: Python, Java, C, JavaScript, SQL **Frameworks:** React, Next.js, Flask, TensorFlow, Face.js, MediaPipe

Platforms: MongoDB, Vercel, Azure, ServiceNow **Tools:** Git, GitHub, CI/CD, Postman, Docker

AI/ML: Machine Learning, Deep Learning, Computer Vision, LLMs, RAG

AI Systems & SaaS: Built AI-as-a-Service (AIaaS) platforms with scalable RAG pipelines, vector databases, secure architecture, and admin dashboards

Projects

Face Recognition — AI-Driven Attendance Automation | attendai.manikantadarapureddy.in

- Engineered an AI-driven, scalable web application for automated attendance using real-time facial recognition.
- Integrated secure authentication, attendance tracking, location validation, and an AI-powered RAG support assistant.
- Achieved **~98–99% facial recognition accuracy** with **~0.2s latency** by optimizing real-time inference and leveraging client-side processing, reducing infrastructure costs by **~$200–$350/month** while improving scalability.
- Deployed and tested in real-world environments with consistent sub-second latency performance.
- Tech: Next.js, **MongoDB**, Face-api.js, **Deep Learning models** (Tiny Face, ResNet, 68-landmark), **RAG Pipeline**.

Vision AI — AI-Powered Visual Detection | visionai.manikantadarapureddy.in

- Developed a real-time visual system for object detection, face landmark tracking, and hand gesture recognition.
- Built a driver drowsiness **detection pipeline** using EAR, PERCLOS, and head pose estimation for safety monitoring.
- Optimized real-time inference for low-latency performance in browser-based environments.
- Integrated a RAG-based support assistant for troubleshooting and user guidance.
- Tech: Next.js, **MediaPipe** (Face Mesh – 468 Landmarks), COCO-SSD, **RAG Pipeline**.

SVECW AI Assistant (RAG) — Intelligent Campus Knowledge Base | svecw.manikantadarapureddy.in

- Built a RAG-based AI assistant with automated document ingestion for context-aware information retrieval.
- Achieved **95–98% response relevance with ~0.5–1s latency** using optimized LLM and vector search pipelines.
- Implemented query-driven recommendations and role-based access control (**RBAC**) for improved user experience.
- Tech: Next.js, OpenAI **GPT-4o-mini**, BAAI/**bge-large-en-v1.5**, **LangChain**, **Pinecone Vector DB**, MongoDB, Clerk Authentication, **Cheerio**, Resend Email Service.

Internships

- Web development – IIT, Academor | **View**
- Full Stack Python Development – APSCHE | **View**
- AWS-CLOUD Computing – AP Govt | **View**

Certifications

- Certified System Administrator – ServiceNow | **View**
- Certified Application Developer – ServiceNow | **View**
- IBM Data Analyst - IBM, Coursera | **View**